Sharper depth dependence for Bayesian networks stable under internal noise

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Abstract

We prove learning bounds for Gaussian Bayesian tanh networks whose clipped predictions are stable under simultaneous internal disturbances. For a layered Boolean teacher of width C and depth $L-1$, a sufficient sample count is $\widetilde{O}(C^5L^{12})$ throughout the mean-field-or-stronger scaling regime $\gamma \geq 1$, independently of learner width above C. At stronger finite scales $\gamma \geq 2$, and in the strong-prior limit at each fixed width, a separate argument gives $\widetilde{O}(C^4L^7)$. Input dimension, robustness calibration, accuracy, and confidence are fixed in these rate displays. No population-merging hypothesis is imposed. The all-scale proof uses a bounded-density property of the conditioned posterior to retain fewer active directions, and represents a Gaussian reconstruction by products of smaller matrices. Rounding during the virtual forward computation makes approximation errors internal disturbances of the uncompressed robust network. We state the earlier reconstruction estimates used and prove the new density and factor-coding arguments. The quartic proof also includes the target-cylinder, sampling, and posterior-tail arguments. These results are conventional proofs, not Lean formalizations; improving the quartic depth power remains open.

1 Introduction

Can an overparametrized Bayesian neural network learn a small circuit with sample complexity polynomial in both circuit width and depth? For tanh networks restricted by stability under internal disturbances, the answer is positive. The question here is how large that polynomial must be. We study exact Gaussian Bayesian prediction conditioned on noiseless training margins, and retain the same calibrated internal-noise support throughout the paper.

Our all-scale result is $\widetilde{O}(C^5L^{12})$ samples, valid at every learner width $N \geq C$ and every mean-field-or-stronger Gaussian scale. A separate stronger-scale argument gives $\widetilde{O}(C^4L^7)$. Both allow any common iid training and test distribution. These are sufficient statistical bounds; neither optimality nor an efficient posterior-sampling algorithm is asserted.

Three ideas give the improved bounds. First, round sampled weights *while constructing* the virtual computation. Each rounding error then becomes a disturbance covered by the original network's stability; the compressed network need not inherit stability. Second, the actual posterior has a bounded density relative to its Gaussian prior. Propagating this bound through finite-message reconstruction gives sharper active-rank tails than a relative-entropy estimate alone. Third, the reconstructed matrices can be stored as products of smaller matrices. A shared Gaussian sketch

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approximates the residual Gaussian field at each fixed input, while keeping a single candidate law for all inputs.

The quartic theorem and its proof come first. They improve the earlier $\widetilde{O}(C^4L^8)$ bound of [1] by one depth power. Appendix B states the imported reconstruction estimates and gives the intermediate sextic bound. Appendix C proves the bounded-density and factor-coding improvements that yield the fifth-power, twelfth-depth result. The target is a layered Boolean circuit; circuit width includes constant wires and compilation overhead as specified next.

2 Learner, target, and robustness restriction

Let $d \geq 1$, $L \geq 2$, and $H = L - 1$. Inputs lie in $\mathcal{X} = \{-1, 1\}^d$, with arbitrary distribution D. A learner of hidden width N has parameters $w = (W^1, \ldots, W^H, v)$ and computes

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}), \qquad z_w(x) = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The raw law $\mu_{N,t}$ has independent centered Gaussian entries of variance t^2/d in W^1, t^2/N in the internal matrices, and t^2/N^2 in v. Here $0 < t \leq 1$. The scale may be any prescribed function $t = t(d, L, N)$; it need not be constant as width varies. In particular $t(N) = N^{(1-\gamma)/2}$ gives variances $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-\gamma-1}$.

Define weighted energy, readout energy, and clipped score by

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N \|v\|_2^2 ,$$
$$E_v(w) = N \|v\|_2^2 , \qquad T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

Disturbances are added after each activation and at the readout:

$$h^1_\eta = \tanh(W^1 \bar{x}) + \eta^1, \quad h^s_\eta = \tanh(W^s h^{s-1}_\eta) + \eta^s, \quad z_w(x; \eta) = v^\top h^H_\eta + \eta^{\text{out}}.$$

Their norm is

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\text{out}}|^2.$$

Fix $c > 0$ before seeing labels. The support condition is

$$\mathcal{R}_{N,c} = \left\{ w : |T(z_w(x; \eta)) - T(z_w(x))| \leq \frac{1}{16} \text{ for all } x \in \mathcal{X}, \quad \|\eta\|_* \leq \frac{c}{\sqrt{1 + HE_v(w)}} \right\}. \tag{1}$$

Noisy hidden values are allowed outside $[-1, 1]^N$. This is a label-independent Borel condition, since the input set is finite and the response is continuous in weights and disturbances.

For a sample $S = ((X_i, f(X_i)))_{i=1}^n$ with iid $X_i \sim D$, let $Q_{S,N,t,c}$ be the raw Gaussian conditioned on

$$w \in \mathcal{R}_{N,c}, \qquad f(X_i) z_w(X_i) > 1/2 \quad (1 \leq i \leq n). \tag{2}$$

There is no initial clean-output-margin condition on unobserved inputs. The risk is posterior-average test-margin failure,

$$R(Q) = \mathbb{E}_{w \sim Q} \mathbb{P}_{X \sim D} \{ f(X) z_w(X) \leq 1/4 \}.$$

The target is a single-output Boolean circuit of width $C \geq 2$ and H layered stages, using NAND, COPY, and NOT. A constant-one wire is reserved at every stage and counted toward C; compilation overhead from another basis is also counted. Unused slots and stages may contain constants or copies. Thus a teacher with $h \leq H$ stages can be padded to the learner's depth. If the learner matches the teacher's depth, $L = h + 1$; if it is deeper, its actual L occurs in the bound. Width independence does not mean depth independence.

3 Learning results

Theorem 1 (All-scale width and depth bound)**.** *Fix $d, c, \varepsilon, \delta$ with $c > 0$ and $0 < \varepsilon, \delta < 1$. There is a constant A, depending only on these four quantities, such that*

$$n \geq AC^5L^{12}[1 + \log^2(2 + C)]^5 \log(2 + CL)$$

implies, with sample probability at least $1 - \delta$, $R(Q_{S,N,t,c}) \leq \varepsilon$ simultaneously for every $N \geq C$ and $0 < t \leq 1$. This holds for every Boolean teacher and common iid input law specified above. The same conclusion holds after additionally restricting the prior to $|z_w(x)| > 1/2$ at every Boolean input.

With all resources variable, a universal constant times $[2+d+L+C+c+c^{-1}+\varepsilon^{-1}+\log(1/\delta)]^{33}$ is also a sufficient sample count.

The exact budget, its statistical dependence, and the proof are in Appendix C, specifically Theorem 7, Corollary 8, and Remark 11. The result applies to the actual finite posterior at every indicated scale. Its proof uses bounded density, not merely a relative-entropy bound on an arbitrary weight distribution.

3.1 A sharper width power at stronger scales

The following separate argument trades its stronger scale hypothesis for quartic dependence on teacher width. Define

$$\begin{aligned}
e_* &= \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = e_*/4, \\
b_L &= \sqrt{16\log 9 + 8\log(4L)}, \\
A &= \max\{32b_L, 2\log(8/e_*)\}, \qquad a = \max\{2, 256c\}, \\
\rho &= \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}, \\
q_0 &= d + 2 + (L-2)C, \\
B_0 &= 9dA^2 + 18(L-2)CA^2 + 2a^2C, \\
D_* &= B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned} \tag{3}$$

These constants provide a target-realizing subset of the prior support of raw mass at least $\exp(-ND_*/t^2)$ at every $N \geq C$; the proof is in Appendix A. In particular $D_* \geq 1$ and every posterior in the theorem below is well-defined.

For $K \geq 1$, set

$$\begin{aligned}
r &= \min\{1/6, c/\sqrt{1+HK}\}, & m &= \left\lceil 256Kr^{-2}[d + \log(400L)] \right\rceil, \\
B &= \sqrt{4mK}, & \zeta &= \frac{r}{4\sqrt{(d+1)^2 + Hm^2}}, \\
P &= m(d+1) + (L-2)m^2 + m, & J &= P\log(3 + 2B/\zeta).
\end{aligned} \tag{4}$$

Theorem 2 (Uniform-width learning with polynomial coding precision)**.** *Let $0 < \varepsilon, \delta < 1$ and suppose f has the Boolean implementation specified above. The sample condition*

$$n \geq \frac{2}{\varepsilon}\left[J + \log(1/\delta)\right] \tag{5}$$

suffices in either of the following cases:

1. *Set $K = 8D_*$. Allow every $N \geq C$ and every positive scale satisfying*

$$t^2 \leq \min\left\{1, \frac{ND_*}{(P_N/2)\log 2 + \log(2/\varepsilon)}\right\}, \qquad P_N = N(d+1) + (L-2)N^2 + N.$$

2. *Set $K = 8D_* + 4(d+L) + 4\log(2/\varepsilon)$. Allow every $N \geq C$ and $0 < t \leq N^{-1/2}$, including every fixed exponent $\gamma \geq 2$.*

With probability at least $1-\delta$ over the sample, the risk $R(Q_{S,N,t,c})$ is at most ε simultaneously for all indicated widths and scales.

For every fixed width, the first clause permits all sufficiently small positive scales and therefore gives the strong-prior risk conclusion $\limsup_{t\downarrow 0} R(Q_{S,N,t,c}) \leq \varepsilon$ on the same sample event. No identification of a separately defined zero-temperature posterior is needed. If σ denotes the strong-prior parameter in the convention with density proportional to $\exp(-\kappa_N/(2\sigma^2))$, then $t = \sqrt{N}\,\sigma$.

Corollary 3 (Explicit width and depth powers)**.** *For fixed $d, c, \varepsilon, \delta$, put*

$$F = 1 + \log^2(2+C) + \log(2+L), \qquad G = \log(2+CL).$$

In either scale regime of Theorem 2, a sufficient sample count has order

$$O\Big(C^4 L^7 F^4 \log^2(2+L) G\Big) = \widetilde{O}(C^4 L^7). \tag{6}$$

At fixed depth this is $O(C^4 \log^9(2+C))$; at fixed width it is $O(L^7 \log^7(2+L))$.

The exact statistical dependence is in (4)–(5). In the second scale regime accuracy also enters K through $\log(2/\varepsilon)$. The asymptotic display (6) holds other resources fixed; it is not an optimality claim.

For the usual confidence criterion, define $p_S(x) = Q_{S,N,t,c}\{f(x)z_w(x) > 1/4\}$. Then $D\{p_S \leq 2/3\} \leq 3R(Q)$; using $\varepsilon/3$ in the theorem gives correct-margin posterior probability strictly greater than $2/3$ on all but an ε fraction of test inputs. This also gives correct-sign confidence and covers arbitrary posterior mixtures.

4 A quantized sampling construction

We prove the deterministic finite-code statement needed for the posterior. Fix N and a network w with $\kappa_N(w) \leq K$. Write $k_s = \|W^s\|_F^2$ for $2 \leq s \leq H$ and $k_L = N\,\|v\|_2^2$. Their sum is at most K.

4.1 Pruning and conditional concentration

In every internal row and in the readout, remove entries with absolute value greater than

$$\frac{8\sqrt{K}\;\|\mathrm{row}\|_2}{r\sqrt{N}}.$$

Zero rows remain zero. Denote pruned arrays by $\bar{W}^s$ and $\bar{v}$. The absolute mass removed from a row is at most its squared norm divided by this threshold. On any vector in $[-1,1]^N$, the internal pruning error therefore has Euclidean norm at most $r\sqrt{Nk_s/K}/8$; the readout error is at most $r\sqrt{k_L/K}/8$.

Let $S^1, \dots, S^H$ be independent lists of m uniform indices in $\{1, \dots, N\}$, with replacement. Let $\mathcal{Q}_\zeta$ round a real number to a nearest multiple of ζ, with a fixed tie convention. It preserves zero and has error at most ζ. Define the *full virtual states* by

$$\begin{aligned}
\widehat{h}_i^1(x) &= \tanh\left(\sum_{j=1}^{d+1} \mathcal{Q}_\zeta(W_{ij}^1)\bar{x}_j\right), \\
\widehat{h}_i^s(x) &= \tanh\left(\sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{W}^s_{i,S_j^{s-1}}\right)\widehat{h}^{s-1}_{S_j^{s-1}}(x)\right), \quad 2 \le s \le H, \\
\widehat{z}(x) &= \sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{v}_{S_j^H}\right)\widehat{h}^H_{S_j^H}(x).
\end{aligned} \tag{7}$$

The stored coordinates at the lists S^s form an ordinary width-m tanh network. This holds even when $m > N$ or indices repeat.

For each $s \ge 2$, the state $\widehat{h}^{s-1}$ depends only on $S^1, \dots, S^{s-2}$. It remains in $[-1,1]^N$ and is independent of the fresh list S^{s-1}. Conditional on the preceding lists, for one uniform index I define

$$Y = N\bar{W}^s_{:,I}\widehat{h}^{s-1}_I.$$

The pruning threshold and bounded states give

$$\mathbb{E}\,\|Y - \mathbb{E}Y\|_2^2 \le Nk_s, \qquad \|Y - \mathbb{E}Y\|_2 \le 16\sqrt{NKk_s}/r.$$

Hilbert-space Bernstein [4], applied at tolerance $r\sqrt{Nk_s/K}/8$, bounds the failure probability by

$$2\exp\left(-\frac{3mr^2}{640K}\right) \le 2\exp\left(-\frac{mr^2}{256K}\right).$$

The readout uses the independent final list S^H and has the same calculation without factors $\sqrt{N}$. Zero-energy stages are exact. A union bound over the 2^d inputs and at most L stages gives simultaneous success probability at least $199/200$ by (4). The concentration estimate is applied *before* coefficient rounding at the current stage, but at the already rounded preceding virtual state. This is why its conditional independence remains valid.

4.2 All errors are disturbances of the original network

On this concentration event, pruning and sampling together give stage preactivation errors at most $r\sqrt{k_s/K}/4$ in normalized Euclidean norm, with the scalar counterpart at the readout. Tanh is 1-Lipschitz, so the corresponding stacked postactivation/readout error has norm at most $r/4$.

Rounding adds at most $(d+1)\zeta$ per first-layer coordinate, and at most $m\zeta$ per coordinate in every later layer and at the readout. This uses only that all virtual hidden states in (7) lie in $[-1,1]^N$. The stacked rounding error thus satisfies

$$\|\eta_{\text{round}}\|_*^2 \le \zeta^2\left[(d+1)^2 + Hm^2\right] = r^2/16.$$

Define the actual disturbances in the original, unpruned network by

$$\eta^1 = \widehat{h}^1 - \tanh(W^1\bar{x}), \qquad \eta^s = \widehat{h}^s - \tanh(W^s\widehat{h}^{s-1}), \qquad \eta^{\text{out}} = \widehat{z} - v^\top\widehat{h}^H.$$

They reproduce the virtual states and output exactly. Splitting each into the pruning/sampling and rounding contributions and using the triangle inequality in the stacked norm gives

$$\|\eta\|_* \le r/4 + r/4 = r/2. \tag{8}$$

If $w \in \mathcal{R}_{N,c}$, then $r \le r_c(w)$ because $E_v(w) \le K$. The original robustness assumption therefore yields

$$\sup_{x \in \mathcal{X}} |T(z_w(x)) - T(\widehat{z}(x))| \le 1/16. \tag{9}$$

There is no subsequent unprotected rounding step, and no assertion that the sampled network itself satisfies (1).

4.3 A finite code independent of learner width

Consider the same sampled coefficients before applying $\mathcal{Q}_\zeta$. Their energy expectations obey

$$\frac{d}{m}\mathbb{E}\left\|W^1_{\mathrm{comp}}\right\|_F^2 = \frac{d}{N}\left\|W^1\right\|_F^2, \qquad \mathbb{E}\left\|W^s_{\mathrm{comp}}\right\|_F^2 \le \|W^s\|_F^2, \qquad m\mathbb{E}\,\|v_{\mathrm{comp}}\|_2^2 \le N\,\|v\|_2^2.$$

These statements concern sampled parameters, not virtual activations. They use independence of adjacent index lists and are unaffected by rounding inside (7). Markov's inequality gives sampled unrounded energy at most $4K$ with probability at least $3/4$. This event intersects the concentration event.

On the intersection, every unrounded stored coefficient has absolute value at most $B = \sqrt{4mK}$. Its rounded value belongs to a fixed grid containing at most $3 + 2B/\zeta$ points. Although rounding was defined on all real numbers for the full virtual computation, only these bounded stored coefficients enter its width-m realization. Thus a fixed class $\mathcal{G}$ of rounded networks satisfies

$$\log|\mathcal{G}| \le P \log(3 + 2B/\zeta) = J, \tag{10}$$

and every internally robust energy-K network at every width has a member of this class satisfying (9).

5 Posterior learning and the sample polynomial

Proof of Theorem 2. Let ψ equal one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and be affine between these thresholds. Every robust network of energy at most K has a rounded representative $g \in \mathcal{G}$. If it fits a training margin, then

$$f(X_i)T(g(X_i)) \ge 1/2 - 1/16 = 7/16 > 5/12.$$

Every test input where $f(x)z_w(x) \le 1/4$ satisfies $f(x)T(g(x)) \le 1/4 + 1/16 = 5/16 < 1/3$. The representative therefore has zero empirical ramp loss, and its population ramp loss bounds the original network's test-margin error.

For a fixed nonnegative loss bounded by one with population mean u, the chance that all n iid losses equal zero is at most $(1-u)^n \le e^{-nu}$. A union bound using (10) and (5) shows that, with probability at least $1-\delta$, every zero-empirical-loss representative has population loss at most $\varepsilon/2$. This single sample event controls every robust fitting network of energy at most K, uniformly over width and scale.

The target cylinder bounds the raw conditioning probability below by e^{-ND_*/t^2} for every correctly labeled sample. Under the raw Gaussian, $N\kappa_N/t^2$ is chi-square with P_N degrees of freedom. Its exponential moment gives

$$Q_{S,N,t,c}\{\kappa_N > K\} \le \exp\Big(-\frac{NK}{4t^2} + \frac{P_N}{2}\log 2 + \frac{ND_*}{t^2}\Big). \tag{11}$$

For $K = 8D_*$, the first scale threshold makes this at most $\varepsilon/2$. For the second choice of K, use $t^2P_N/N \le d + L$ when $t^2 \le 1/N$. The exponent multiplied by t^2/N is at most

$$-K/4 + D_* + (d+L)(\log 2)/2 \le -\log(2/\varepsilon).$$

Since $N/t^2 \ge 1$, the tail is again at most $\varepsilon/2$. Adding this discarded posterior mass to the retained risk proves the claim for arbitrary posterior mixtures. □

Proof of Corollary 3. Put $T_L = d + \log(400L)$. The budget gives $m = O(Kr^{-2}T_L)$. All factors in B/ζ are polynomial in d, L, K, r^{-1}, so

$$J = O\Big([dKr^{-2}T_L + LK^2r^{-4}T_L^2]\log(2 + d + L + K + r^{-1})\Big).$$

The target constants imply

$$D_* = O\Big((d + LC)[\log^2(2C) + \log(2L) + \log(2 + d + C)] + (1 + c^2)C\Big).$$

At fixed $d, c, \varepsilon, \delta$, it follows that

$$K = O(CLF), \quad r^{-2} = O(CL^2F), \quad m = O(C^2L^3F^2\log(2 + L)),$$
$$P = O(C^4L^7F^4\log^2(2 + L)), \qquad \log(3 + 2B/\zeta) = O(G).$$

Substitution proves (6) and its fixed-width and fixed-depth consequences. □

6 Scope and remaining losses

The argument retains precisely the original internal-noise restriction. The improvement is in its finite code: rounding errors are absorbed during sampling rather than propagated afterward through the compressed network. The latter propagation required a log grid cost of order $L\log(2+d+L+K+r^{-1})$; here it has no leading factor of L.

The remaining depth powers come from the target cost, the calibrated noise radius, sampling width, and dense parameter counting. In particular, m is of order K/r^2 up to logarithms, and the code has order Lm^2 parameters. The posterior proof bounds readout energy by total energy, which can be of order CL up to logarithms. Removing more depth factors would require improving these estimates or using a different argument. No matching lower bound is asserted.

At mean-field scaling, expected internal Frobenius energy grows with width; for $1 < \gamma < 2$, it grows as $N^{2-\gamma}$. Thus the total-energy tail used here does not prove the quartic rate in that interval. Other compression methods give a polynomial learning theorem there [3], but that is a separate argument. No stronger support condition is introduced to bypass this limitation in the present theorem.

The target-cylinder and sampling ingredients build on the robust-prior and earlier quartic papers [2, 1], including their collaborative contributions. The proof presented here is conventional, not Lean formalized. Formal proofs for other papers in this research program do not certify this revised quantized sampling construction.

A A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \geq C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \leq N/k_a \leq 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16 \log 9 + 8 \log(4L)},$$
$$A = \max\{32 b_L, 2 \log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], A e_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\, W^1$, $\sqrt{N}\, W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t \sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A $1/4$-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2 \exp(2N \log 9 - N b_L^2/8) \leq \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least $1/2$.

Finite disturbance robustness. Allow a separate RMS disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \leq \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \leq e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \leq 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \geq A/2 \quad \Longrightarrow \quad |\tanh(q + u) - \epsilon| \leq 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h^{s-1}_\eta\|_2/\sqrt{N} \leq 1 + \tau_0$, it follows that

$$e_s \leq 2e^{-A/2} + (8/A)\rho\sqrt{C}(1 + \tau_0) + (8b_L/A)e_{s-1} + \tau_0 \leq e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \leq A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x) z_w(x; \eta) \geq \frac{15a}{16} - \rho(1 + \tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \leq \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \geq a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v-v_*\| \leq \rho$. Therefore $E_v \geq a^2C/8$, whence

$$r_c(w) \leq \frac{2\sqrt{2}c}{a\sqrt{C}} \leq \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \leq 1/4$, and any center m,

$$\Pr\{|Z-m| \leq \rho t\} \geq \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \geq \exp(-ND_*/t^2), \qquad D_* = B_0 + q\log\frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{12}$$

Here $D_* \geq 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \geq C$.

B Sequential quantization at every mean-field-or-stronger scale

The deterministic compressor used in the main theorem requires a bound on total network energy. At the finite mean-field scale, the unconditioned Gaussian background can have growing total energy. An entropy-based compressor avoids that difficulty. In this appendix we combine the same sequential-quantization principle with the active-direction and covariance reconstruction estimates of [3]. The result improves the unchanged internal-noise prior at every $0 < t \leq 1$, including $\gamma = 1$. This appendix is not a new proof of those active-direction estimates: we state precisely which properties are imported and prove the modification that removes the coding precision's extra depth factor. Neither this modification nor its new depth exponent is Lean formalized.

B.1 Budgets and learning statement

Use the model, support $\mathcal{R}_{N,c}$, and posterior $Q_{S,N,t,c}$ from (1)–(2). Let D_* be any target-cylinder cost satisfying the conclusion of Appendix A. Thus the raw posterior conditioning event has probability at least $\exp(-ND_*/t^2)$, for every correctly labeled sample and every $N \geq C$, $0 < t \leq 1$. For $0 < \varepsilon, \delta < 1$, put

$$\beta = \varepsilon/4, \qquad B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, \qquad K = D_* + 1,$$
$$r = \min\{1/6, c/\sqrt{1 + (L-1)B_v}\}, \qquad \alpha = \varepsilon/12.$$

Define the compression parameters

$$
\begin{aligned}
e &= \frac{\alpha r^2}{1000L}, & \eta &= \frac{\alpha}{100L}, & q_a &= \left\lceil \frac{2K}{e\eta} \right\rceil, \\
D_0 &= [(d+1)q_a + 2]\log 2, & V &= 8K + 2q_a, & B_H^2 &= 4V/e, \\
K_0 &= (4K + 2\log 2)/\eta, & B_1^2 &= 4[4K + 2(d+1)\log 2]/e.
\end{aligned}
$$

The letter q_a denotes the number of retained active directions; it is unrelated to a population partition or a support restriction. Let m be the least integer no smaller than each of

$$
\begin{aligned}
&10000(V + q_a + 1)/e, \\
&64(q_a + 1)\log(4(q_a + 1)/\eta), \\
&512K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1].
\end{aligned}
$$

Finally set

$$
\begin{aligned}
M &= \max\{m, \lceil LD_0/K \rceil, 1\}, & B_G &= \max\{2, 2\sqrt{\log(6m/e)}\}, \\
P_* &= M(d+1) + (L-2)M^2 + M, & B_{\text{small}}^2 &= \frac{4(4MK + 2P_*\log 2)}{\alpha}, \\
B_{\text{large}} &= \max\left\{B_1, \frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}, \frac{\sqrt{K_0}}{m}, 1\right\}, \\
B_* &= 1 + \max\{B_{\text{small}}, B_{\text{large}}, 1\}, & \zeta &= \frac{\sqrt{e}}{4(M+d+1)}, \\
J_* &= P_*\log(3 + 4B_*/\zeta).
\end{aligned}
$$

All square roots above denote the nonnegative root.

Theorem 4 (An improved all-scale budget)**.** *Under the Boolean-teacher convention of the main theorem, the condition*

$$
n \geq \frac{16}{\varepsilon}\left[J_* + \log(1/\delta)\right] \tag{13}
$$

implies, with probability at least $1 - \delta$ over iid training inputs,

$$
\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon
$$

simultaneously for every $N \geq C$ and every $0 < t \leq 1$. The conclusion is also valid if global clean-output-margin conditioning is added to the prior. No population-merging or additional robustness assumption is made.

With $d, c, \varepsilon, \delta$ fixed, write

$$
F = 1 + \log^2(2 + C) + \log(2 + L), \qquad G = \log(2 + CL).
$$

The target constants and the budgets above give

$$
n = O(C^6 L^{17} F^6 G). \tag{14}
$$

This improves the former $O(C^6 L^{18} F^6 G)$ budget while preserving its fixed-depth $O(C^6 \log^{13}(2 + C))$ width consequence. A further target-cylinder sharpening below removes the $\log(2 + L)$ term from F.

B.2 Imported entropy and reconstruction estimates

We use the following properties of the construction in the appendix "Finite posterior compression by internal-noise coupling" of [3]. Its lemmas on future-weight entropy, whole-layer active directions, and whitened sampling prove these properties for *any bounded virtual prefix*. They do not require a Gaussian law for the prefix and do not assume its robustness.

First, suppose a prefix is A, the current original weight block is X, and the future blocks are Y, with product Gaussian reference law $P_X \otimes P_Y$. For a finite message J, a retained quantity $H_0 = f(A, J, X)$, and a new prefix A' which is a randomized function of (A, J, H_0) using fresh randomness conditionally independent of Y,

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'} \| P_Y) \le \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A} \| P_X \otimes P_Y) + \mathcal{H}(J \mid A). \tag{15}$$

Here $\mathcal{H}$ denotes conditional Shannon entropy. The inequality is the KL chain rule and data processing; deterministic processing of A' cannot increase its left side.

Second, if $Q \ll \mu_{N,t}$ has $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$ and $N > M$, the budgets above permit the following sequential construction. Its current virtual layer has N rows, each evaluated on m stored source neurons from the preceding layer. The first original layer is clipped rowwise at physical standardized norm B_1. Each internal stage retains at most q_a active input directions, clips the active row norms at B_H, reconstructs the complementary covariance by whitened source sampling, and clips its new Gaussian coefficients at B_G. The finite message selecting active inputs has entropy at most D_0; all other new randomness is fresh conditional on the prefix and active matrix. Hence (15) bounds the unprocessed original-weight entropy at stage s by

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2.$$

For every fixed input x, the conditional reconstruction couplings give expected squared RMS activation injection less than $20e$ per hidden stage, with success indicators. The first layer has the smaller bound e. Before any later row selection, each new internal coefficient has absolute value at most

$$\frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}.$$

The readout is truncated at $N\|v\|^2 \le K_0$ and reconstructed using m source samples, with coefficient bound $\sqrt{K_0}/m$ and uniform output error at most $r/16$. The sum of probabilities of rank, Gram, and readout failures is at most $3L\eta$.

For clarity, the quantitative source-reconstruction input is the following estimate. For bounded source activations $|h_j(x)| \le 1$, active dimension at most q_a, and active matrix H_0, its whitened half-uniform, half-leverage sampling law has expected squared RMS error, with the good-Gram indicator, at most

$$\frac{2\|H_0\|_F^2/N + 16(q_a+1)}{m}.$$

The averaged active energy is bounded by V, while the Gram failure probability is at most η. These facts, Gaussian clipping, the active-direction transport estimate, and the squared triangle inequality give the stated $20e$ bound. Each bound holds for every bounded prefix to which the entropy estimate applies. These are the nontrivial reconstruction estimates imported from [3]; the argument below does not infer them merely from an existing compressed network's clean-output accuracy.

B.3 Rounding inside the canonical construction

Choose fixed scalar lattice rounding with error at most ζ and with zero fixed. After first-layer clipping, round all its coefficients before constructing any later prefix. At each internal stage apply the imported reconstruction procedure to the *already rounded* prefix, then round every newly constructed virtual row before processing the next original layer. Finally round the reconstructed readout. Keep the source rows selected at subsequent stages, as in the unrounded construction. There is no terminal rounding pass.

This modification preserves the preceding estimates. Tanh keeps the rounded virtual activations in $[-1, 1]$. Each preceding state is specified before the next fresh source list is sampled. Thus the boundedness and conditional-independence assumptions of the sampling estimate remain valid. Rounding new rows is a deterministic function of the same prefix, message, active matrix and fresh Gaussian variables as their unrounded construction. It therefore preserves (15) without adding a message. The resulting prefix may have a discrete law; that law never receives a Gaussian relative-entropy comparison.

Fix an input x. The conditional transports can be glued while keeping the original full-network marginal equal to Q and the outgoing rounded-prefix/future-weight marginal equal to the next canonical stage's law. To do so, extend each transition by the previous conditional law of past original weights given its incoming prefix and remaining original weights. This disintegration preserves all earlier computed prefixes. Define the new postactivation injections by

$$\eta^s = h^s_{\text{rounded virtual}}(x) - \tanh\big(W^s_{\text{original}} h^{s-1}_{\text{rounded virtual}}(x)\big).$$

The final candidate output is exactly the original network's output under these injections and the analogous readout injection. Its marginal law is independent of x, although the coupling may depend on x.

Coefficient rounding changes a first-layer preactivation by at most $(d+1)\zeta$, and every later virtual preactivation by at most $m\zeta$. Both are at most $\sqrt{e}/4$. Consequently the expected squared RMS hidden injection, including rounding, is at most

$$2(20e) + 2(e/16) < 41e.$$

Outside the abort events, Markov's inequality gives

$$\mathbb{P}\left\{\sum_{s<L} \|\eta^s\|^2 / N > r^2/16\right\} \le \frac{656Le}{r^2} = 0.656\alpha.$$

The total abort probability is at most $3L\eta = 0.03\alpha$. The readout rounding error is at most $m\zeta \le \sqrt{e}/4 \le r/64$, where the last inequality uses $\alpha \le 1/8$ and $L \ge 2$. Together with its sampling error, the scalar injection is at most $5r/64$. On the good event,

$$\|\eta\|_* \le \sqrt{r^2/16 + 25r^2/4096} = \frac{\sqrt{281}}{64} r < r/2.$$

The exceptional probability is less than α.

When $N \le M$, the Gaussian entropy-moment inequality instead gives, with probability at least $1 - \alpha/4$, the coefficient bound B_{small} above. On this event round every original coefficient and compute the rounded network sequentially; on its complement use a zero-readout default. At each input the rounded computation is the original network under injections satisfying

$$\|\eta\|_*^2 \le \zeta^2[(d+1)^2 + (L-1)N^2] \le \zeta^2 L(M+d+1)^2 = Le/16 < r^2/4.$$

This identity uses bounded rounded hidden activations, not a bound on products of original matrix norms.

All retained coefficients, after rounding, lie in $[-B_*, B_*]$. Padding each candidate to width M with zero neurons is exact for tanh. A grid with at most $3 + 4B_*/\zeta$ entries per coefficient gives a finite class $\mathcal{H}_*$ with

$$\log |\mathcal{H}_*| \le J_*. \tag{16}$$

The construction proves the following strengthened finite-code statement: for every such Q there is one law Γ_Q on $\mathcal{H}_*$ such that, at each fixed input, a coupling has

$$h(x) = z_w(x; \eta), \qquad \|\eta\|_* < r/2$$

except on probability at most α. Only the coupling, not its two marginals, can depend on the input.

B.4 Posterior and statistical argument

Under the raw Gaussian, NE_v/t^2 is chi-square with N degrees of freedom. The target-cylinder denominator and the exponential moment bound give posterior mass at most β to $E_v > B_v$. The retained conditional posterior Q' satisfies

$$D_{\mathrm{KL}}(Q' \| \mu_{N,t}) \le ND_*/t^2 + \log(1/(1-\beta)) \le NK/t^2.$$

Every retained original network is robust on the common radius r. Apply the strengthened finite-code statement to Q'. On each good coupling the clipped-score error is at most $1/16$.

Let ψ be one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and affine between these intervals. A correctly fitted training point has candidate ramp loss zero on the good coupling. A test-margin failure of the original network has candidate ramp loss one there. For the finite class $\mathcal{H}_*$, a relative Chernoff inequality and a union bound give, on a sample event of probability at least $1 - \delta$,

$$\mathbb{E}_D \psi(f(X)T(h(X))) \le \frac{2}{n} \sum_{i=1}^{n} \psi(f(X_i)T(h(X_i))) + \frac{8[J_* + \log(1/\delta)]}{n} \qquad (h \in \mathcal{H}_*).$$

Averaging over $\Gamma_{Q'}$, comparing population risks, and restoring the discarded posterior mass bounds the desired risk by

$$\beta + 3\alpha + \frac{8[J_* + \log(1/\delta)]}{n} \le \varepsilon$$

under (13). The finite class and sample event do not depend on N, t. This proves Theorem 4. Adding global-margin conditioning changes none of these steps, since the same target cylinder is retained.

B.5 Width–depth accounting and a sharper target cylinder

For fixed $d, c, \varepsilon, \delta$, the displayed budgets give

$$\begin{aligned} D_*, K, B_v &= O(CLF), & r^{-2} &= O(CL^2F), \\ e^{-1} &= O(CL^3F), & \eta^{-1} &= O(L), \\ q_a, V &= O(C^2L^5F^2), & M &= O(C^3L^8F^3), \\ P_* &= O(C^6L^{17}F^6). \end{aligned}$$

Both B_* and ζ^{-1} are polynomial in C, L, F, so $\log(3 + 4B_*/\zeta) = O(G)$. The old terminal grid had an additional factor L here. This establishes (14).

There is a separate improvement to the target-cylinder budget which changes neither the prior nor the compression. In Appendix A, replace the depth-dependent bulk cutoff by

$$b = \sqrt{16 \log 9 + 8 \log 4}, \qquad A = \max\{32b, 2 \log(8/e_*)\}.$$

Each unconstrained Gaussian bulk matrix has, by the same two-net bound,

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > bt\} \le 2 \exp(2N \log 9 - Nb^2/8) = 2 \cdot 4^{-N} \le 1/2.$$

The bulk matrices are conditionally independent across layers. Their simultaneous success probability is therefore at least $2^{-(H-1)}$. Every deterministic stability estimate in the cylinder proof is unchanged. Recompute B_0, q_0, ρ with this constant-cutoff choice of A and use the conservative cost

$$D_*^{\mathrm{prod}} = B_0 + q_0 \log \frac{e\sqrt{2\pi}}{2\rho} + H \log 2. \tag{17}$$

Indeed $N/t^2 \ge 1$ absorbs the product-event cost into this rate. At fixed d, c, with $F_0 = 1 + \log^2(2 + C)$, one has $D_*^{\mathrm{prod}} = O(CLF_0)$. Substituting this valid cylinder cost in Theorem 4 gives

Corollary 5 (Depth logarithms removed from the target gain)**.** *For fixed $d, c, \varepsilon, \delta$, the unchanged robust prior admits sufficient sample count*

$$O\Big(C^6 L^{17} [1 + \log^2(2 + C)]^6 \log(2 + CL)\Big)$$

at every $N \ge C$ and every $0 < t \le 1$. At fixed teacher width this is $O(L^{17} \log(2 + L))$.

The same cylinder replacement is also available in the quartic main theorem; it is independent of the sampling and quantization argument. Its proof uses the target constants only through their valid mass lower bound. With this substitution the quartic sufficient order becomes

$$O\Big(C^4 L^7 [1 + \log^2(2 + C)]^4 \log^2(2 + L) \log(2 + CL)\Big).$$

At fixed teacher width this is $O(L^7 \log^3(2 + L))$.

Remark 6 (A conservative joint polynomial)**.** *Put $S = 2 + d + L + C + c + c^{-1} + \varepsilon^{-1} + \log(1/\delta)$. The coarse accounting in [3] for the unchanged budgets gives $e^{-1} = O(S^8)$, $M = O(S^{21})$, $P_* = O(S^{43})$, and $B_* = O(S^{22})$. The new mesh has $\zeta^{-1} = O(S^{25})$. Thus $J_* = O(S^{43} \log S)$ and (13) is $O(S^{44} \log S)$, hence $O(S^{45})$. This deliberately loose degree-45 envelope is a consequence of the conventional modification proved here. It does not replace or enlarge the existing Lean certification of the earlier degree-46 envelope.*

C Bounded-density compression with factored Gaussian reconstruction

The all-scale code in Appendix B can be sharpened in two further ways. First, the actual conditioned posterior has a bounded density relative to the raw Gaussian. Propagating this stronger property through the virtual computation removes the factor $1/\eta$ from the active-direction cutoff. This argument is specific to bounded density; it does not upgrade the compression theorem for arbitrary

laws having only a relative-entropy bound. Second, the active part of each reconstructed matrix already has small rank, and its fresh Gaussian part can be replaced by a small-rank Gaussian factorization. Coding the factors then gives a sufficient sample order $\widetilde{O}(C^5L^{12})$ at every $0 < t \le 1$. The support remains exactly $\mathcal{R}_{N,c}$.

We prove the new density and factorization arguments below. We use the whole-layer active-direction and whitened-sampling estimates stated in Appendix B, with their original proofs in [3]. As in that appendix, every comparison coupling is allowed to depend on the input, whereas the candidate law is not. The argument and the improved exponents are conventional proofs and are not Lean formalized.

C.1 Explicit budgets and learning statement

Use the product-cylinder cost $D_* = D_*^{\mathrm{prod}}$ from (17). For $0 < \varepsilon, \delta < 1$, define

$$\begin{aligned} &\beta = \varepsilon/4, && B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, && K = D_* + 1, \\ &r = \min\{1/6, c/\sqrt{1 + (L-1)B_v}\}, && \alpha = \varepsilon/12, && e = \frac{\alpha r^2}{1000L}, \qquad \eta = \frac{\alpha}{100L}. \end{aligned} \tag{18}$$

Set

$$\begin{aligned} &q = \left\lceil \frac{2K + \log(1/\eta)}{e} \right\rceil, && D_0 = [(d+1)q + 2]\log 2, && V = 8K + 2q, \\ &B_H^2 = 4V/e, && K_0 = (4K + 2\log 2)/\eta, && B_1^2 = 4[4K + 2(d+1)\log 2]/e. \end{aligned} \tag{19}$$

Let m be the least integer no smaller than each of

$$\begin{aligned} &10000(V + q + 1)/e, \\ &64(q+1)\log(4(q+1)/\eta), \\ &512K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1]. \end{aligned} \tag{20}$$

The factor rank and the small-width cutoff are

$$p = \lceil 1000/e \rceil, \qquad k = q + p, \qquad n_0 = \max\{1, \lceil LD_0/K \rceil\}. \tag{21}$$

In particular, the cutoff n_0 need not be as large as m. Define factor bounds and parameter counts by

$$\begin{aligned} B_g^2 &= 4p/e, \qquad A_f = \max\{B_H, B_g, 1\}, \\ B_f &= \max\{2\sqrt{q+1}/m, \sqrt{2/\eta}, 1\}, \\ P_{\mathrm{small}} &= n_0(d+1) + (L-2)n_0^2 + n_0, \\ P_{\mathrm{fac}} &= m(d+1) + 2(L-2)mk + m, \\ B_{\mathrm{small}}^2 &= \frac{4(4n_0K + 2P_{\mathrm{small}}\log 2)}{\alpha}, \\ B_{\mathrm{code}} &= 1 + \max\{B_{\mathrm{small}}, B_1, A_f, B_f, \sqrt{K_0}/m, 1\}. \end{aligned} \tag{22}$$

Finally, choose the common scalar mesh

$$\zeta = \min\left\{1, \frac{\sqrt{e}}{4(d+1)}, \frac{\sqrt{e}}{4m}, \frac{\sqrt{e}}{4mk(A_f + B_f + 1)}, \frac{\sqrt{e}}{4(n_0 + d + 1)}\right\}, \tag{23}$$

and put

$$J_{\mathrm{fac}} = \log 2 + \max\{P_{\mathrm{small}}, P_{\mathrm{fac}}\} \log(3 + 4B_{\mathrm{code}}/\zeta). \tag{24}$$

All these quantities are independent of learner width and Gaussian scale.

Theorem 7 (All-scale learning by a factored code)**.** *Under the Boolean-teacher convention of the main theorem, the condition*

$$n \geq \frac{16}{\varepsilon}[J_{\mathrm{fac}} + \log(1/\delta)] \tag{25}$$

implies, with probability at least $1 - \delta$ over iid training inputs,

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ and every $0 < t \leq 1$. The same conclusion holds if the posterior conditioning additionally requires the label-independent global margin $|z_w(x)| > 1/2$ for every input x. No population-merging assumption or additional internal-noise condition is imposed.

Corollary 8 (Improved width and depth powers)**.** *For fixed $d, c, \varepsilon, \delta$, let $F_0 = 1 + \log^2(2 + C)$. A sufficient sample count in Theorem 7 has order*

$$O\Big(C^5 L^{12} F_0^5 \log(2 + CL)\Big) = \widetilde{O}(C^5 L^{12}). \tag{26}$$

At fixed teacher width this is $O(L^{12} \log(2 + L))$; at fixed depth it is $O(C^5 \log^{11}(2 + C))$.

C.2 A bounded-density invariant for unprocessed weights

The following lemma strengthens the entropy bookkeeping at the point where an active rank must hold with high probability. A measure inequality means domination on every measurable set.

Lemma 9 (Conditional density and finite-message propagation)**.** *Suppose that*

$$Q_{A,X,Y} \leq e^b (\nu_A \otimes P_X \otimes P_Y),$$

where ν_A, P_X, P_Y are probability reference laws. Write $u = dQ_{A,Y}/d(\nu_A \otimes P_Y)$. Then

$$Q\{u(A, Y) < \eta\} \leq \eta,$$

and, outside this event,

$$D_{\mathrm{KL}}(Q_{X|A,Y} \| P_X) \leq b + \log(1/\eta). \tag{27}$$

Suppose also that $J = j(A, Y)$ takes values in a fixed finite alphabet $\mathcal{J}$, that $H_0 = h(A, J, X)$, and that a new prefix A' is generated from (A, J, H_0) through a kernel whose fresh randomness has no further dependence on Y. There is a probability reference law $\nu_{A'}$ such that

$$Q_{A',Y} \leq e^{b + \log|\mathcal{J}|} (\nu_{A'} \otimes P_Y). \tag{28}$$

The original domination also implies

$$\mathbb{E}_{Q_A} D_{\mathrm{KL}}(Q_{X,Y|A} \| P_X \otimes P_Y) \leq b.$$

Proof. Let $F \le e^b$ be the joint density relative to $\nu_A \otimes P_X \otimes P_Y$. Disintegration gives conditional density $F(a,x,y)/u(a,y)$ for $Q_{X|a,y}$ wherever $u > 0$. Thus its relative entropy is at most $b - \log u(a,y)$. Also

$$Q\{u < \eta\} = \int_{\{u<\eta\}} u \, d(\nu_A \otimes P_Y) \le \eta.$$

This is an unconditional probability bound under the canonical joint law; it need not hold separately at every fixed prefix.

For propagation, augment the reference law by an independent uniformly distributed message in $\mathcal{J}$. The actual augmented law satisfies

$$Q_{A,J,X,Y} \le e^b |\mathcal{J}| (\nu_A \otimes \mathrm{Unif}(\mathcal{J}) \otimes P_X \otimes P_Y).$$

Apply the retained-data map and the same prefix kernel to both sides. Under the reference law, its inputs and fresh randomness are independent of Y, so its pushforward factors as $\nu_{A'} \otimes P_Y$. This proves (28). The last assertion follows from the KL chain rule and the nonnegativity of $D_{\mathrm{KL}}(Q_A \| \nu_A)$. □

Suppose initially that a law Q on original weights satisfies

$$Q \le \exp(NK/t^2)\, \mu_{N,t}. \tag{29}$$

The virtual prefix records the constructed network, not just its value at one selected input. Clipping and rounding the first layer are deterministic channels applied to that original block and initialize the domination invariant with exponent NK/t^2. At an internal stage, let X be the current original block and Y the remaining original blocks. Induction with Lemma 9 gives exponent

$$b_s \le NK/t^2 + sD_0 \le 2NK/t^2 \qquad (N > n_0).$$

At that stage abort if its marginal density $u(A,Y)$ is below η. Otherwise (27) and the imported whole-layer active-direction lemma, applied with tolerance e, give

$$\dim U \le \frac{t^2}{Ne}[b_s + \log(1/\eta)] \le \frac{2K + \log(1/\eta)}{e} \le q.$$

The active query list is selected from the conditional law given (A,Y), so it is a function of (A,Y). Its basis is a function of the prefix and the list, using fixed Gram–Schmidt conventions. The selected list and an abort symbol admit a fixed alphabet of log-cardinality at most D_0. For reference messages never chosen by the actual selection rule, extend the basis and transition kernel by fixed measurable default rules. This makes the uniform-message reference construction in the lemma defined on its entire domain.

All abort paths remain in the canonical probability law. On an abort, use a fixed default prefix, retain an unsuccessful flag, and use default transitions thereafter. Do not condition any intermediate law on success. The density-test abort is included in the finite message; later aborts arising from fresh randomness are part of the prefix channel. Consequently the invariant applies on the full law, and the unsuccessful paths are discarded only in the final approximation probability estimate.

The domination invariant also provides the old averaged entropy bound. Thus the imported Gaussian entropy-moment estimate still gives

$$\mathbb{E}\left[\|H_0\|_F^2/N;\ \text{active success}\right] \le 4t^2 b_s/N + 2t^2 q \log 2 \le V. \tag{30}$$

In particular, the definitions of V, B_H, B_1, and K_0 need no extra factor of $1/\eta$. The improvement in q uses the density hypothesis (29); an arbitrary KL bound alone would not justify it.

For comparison, using this new rank cutoff with the dense sequential code of Appendix B already gives $O(C^6 L^{15} F_0^6 \log(2 + CL))$ samples. The factor construction below saves additional powers of both resources.

C.3 A small-rank replacement for the Gaussian field

We recall the source-sampling notation needed for the new step. The source activations obey $|h_j(x)| \le 1$. If U has $\ell \le q$ orthonormal columns, put $u_j = \sqrt{N} U_{j,:}$. The half-uniform, half-leverage law gives weights $a_j = 1/(Np_j) \le 2$ and $a_j \|u_j\|^2 \le 2\ell$. Draw m source indices J_b, and define

$$\widehat{G} = \frac{1}{m} \sum_b a_{J_b} u_{J_b} u_{J_b}^\top, \qquad D_{b,:} = \sqrt{a_{J_b}/m}\, u_{J_b}^\top \widehat{G}^{-1/2}.$$

On the good Gram event, $\widehat{G} \succeq I/2$ and $D^\top D = I$. For $\ell = 0$, the active matrices are empty and the following projection is the identity. Write

$$P = I - DD^\top, \qquad c_b(x) = \sqrt{a_{J_b}/m}\, h_{J_b}(x), \qquad \widehat{C}(x,x) = \|Pc(x)\|^2 \le 2.$$

The ideal field in the imported whitened-sampling estimate has active mean $H_0\psi(x)$ and independent Gaussian rows of variance $t^2 C(x,x)$. The original reconstruction has mean $H_0\widehat{v}(x)$ and variance $t^2\widehat{C}(x,x)$.

Lemma 10 (Factored Gaussian reconstruction)**.** *Conditional on the source samples and retained active data, draw an independent standard Gaussian matrix $R \in \mathbb{R}^{p\times m}$ and independent standard Gaussian row vectors $g_i \in \mathbb{R}^p$, for all virtual target rows. Replace the reconstructed coefficients by*

$$\widetilde{W}_{ib} = \frac{a_{J_b}}{m} H_{0,i} \cdot u_{J_b} + t\sqrt{\frac{a_{J_b}}{mp}}\, (g_i^\top RP)_b. \tag{31}$$

This coefficient law does not depend on the input. At every fixed input it can be coupled to the ideal field so that the expected squared RMS evaluation error, with the good Gram indicator, is at most

$$\frac{4\|H_0\|_F^2/N + 32(q+1)}{m} + \frac{8}{p}. \tag{32}$$

The coefficient matrix in (31) has rank at most $\ell + p$.

Proof. Fix the input and abbreviate $\sigma^2 = \widehat{C}(x,x)$. Conditional on R, the new Gaussian evaluations are independent across rows with variance $t^2 s^2$, where

$$s = \|RPc(x)\|/\sqrt{p}.$$

If $\sigma > 0$, then s^2/σ^2 has the law of a chi-square variable with p degrees of freedom divided by p. Rationalizing the square root therefore gives

$$\mathbb{E}_R(s-\sigma)^2 \le \sigma^2 \mathbb{E}(\chi_p^2/p - 1)^2 = 2\sigma^2/p \le 4/p. \tag{33}$$

For $\sigma = 0$ the statement is immediate.

To make the coupling explicit, draw R and independent standard normals Z_i, independent of R. Use Gaussian row evaluations $t\sigma Z_i$ for the old reconstruction and tsZ_i for the new one. Disintegrating g_i conditional on its projection along $RPc(x)$ realizes the latter evaluation while keeping its full conditional marginal standard Gaussian. Thus the coefficient law is exactly the law in (31). Disintegration likewise extends the old Gaussian evaluation to its coefficient field. The degenerate zero-variance case is handled by independent completion. Only this comparison coupling, not either field marginal, depends on x.

The old whitened reconstruction estimate is $[2\|H_0\|_F^2/N + 16(q+1)]/m$. Couple through that reconstruction and apply the squared triangle inequality with (33). This yields (32). Finally the active term factors through $\mathbb{R}^\ell$ and the Gaussian term through $\mathbb{R}^p$, proving the rank claim. □

Integrating (32) with the success indicator and using (30), the first line of (20), and $p \geq 1000/e$, bounds this error by

$$\frac{4V + 32(q+1)}{m} + \frac{8}{p} < e/8.$$

The comparison requires no simultaneous coupling of all inputs. Conditional independence of rows given R suffices; independence of the new rows after averaging over R is not asserted or needed.

C.4 Bounded factors and their sequential rounding

At each internal stage first clip active rows of norm greater than B_H to zero, exactly as in the imported construction. Equation (30) and bounded activations give squared RMS activation cost at most $4V/B_H^2 = e$. Apply Lemma 10 to this clipped matrix.

Abort if $\|R\|_F^2 > pm/\eta$. Since $\mathbb{E}\,\|R\|_F^2 = pm$, this event has conditional probability at most η. Also replace each g_i by zero when $\|g_i\|^2 > B_g^2 = 4p/e$. This row replacement has probability at most $e/4$ and hence expected squared activation change at most e: two tanh values differ by at most two. All virtual rows are clipped before any later row selection, including selections depending on future weights.

On success, write the resulting coefficients as $A_i B$, with zero padding to factor dimension $k = q + p$, where

$$A_i = (H_{0,i}, g_i), \qquad B_{\text{active},ab} = a_{J_b} u_{J_b,a}/m,$$

$$B_{\text{Gaussian},ab} = t\sqrt{a_{J_b}/(mp)}\,(RP)_{ab}.$$

Every left factor entry has absolute value at most A_f. Every right factor entry has absolute value at most B_f, since

$$|a_{J_b} u_{J_b,a}/m| \leq 2\sqrt{q+1}/m, \qquad t\sqrt{\frac{a_{J_b}}{mp}}|(RP)_{ab}| \leq \sqrt{\frac{2}{mp}}\,\|R\|_F \leq \sqrt{2/\eta}.$$

Round both factor arrays to the common scalar mesh (23) before computing the next virtual prefix. For each bounded incoming state,

$$|(A_{i,\text{round}} B_{\text{round}} - A_i B)h| \leq mk(A_f + B_f + 1)\zeta \leq \sqrt{e}/4. \tag{34}$$

Here each product changes by at most $(A_f + B_f + 1)\zeta$, because $\zeta \leq 1$. Round first-layer coefficients and reconstructed readout coefficients directly. Their preactivation errors are at most $(d+1)\zeta$ and $m\zeta$, respectively, again at most $\sqrt{e}/4$. The rounded candidate is an ordinary depth-L tanh network: its internal matrix is the product of two rounded factor arrays. Factorization adds no activation layer.

All source samples, R, and Gaussian row factors are fresh randomness conditional on the prefix, query list, and retained active matrix. Clipping, factor multiplication, and rounding are deterministic subsequent operations. They therefore form a channel of the precise kind required by Lemma 9. The reference-prefix law need not be Gaussian or have a density on its own coordinate space. The comparison coupling does not replace this canonical channel in the density induction.

C.5 The disturbance budget and the finite class

Consider first $N > n_0$ and a law satisfying (29). The first layer is clipped at the physical standardized row bound B_1 and then rounded. Its expected squared RMS activation injection is less than $3e$ by the imported first-layer estimate and the squared triangle inequality. At an internal stage the

five contributions are active-direction transport, active-row clipping, reconstruction, Gaussian-row clipping, and factor rounding. Their expected squared RMS bounds, with success indicators, are respectively

$$2e, \quad e, \quad e/8, \quad e, \quad e/16.$$

Another squared triangle inequality bounds their sum by $5(2+1+1/8+1+1/16)e < 21e$. In particular the conservative bound $41e$ from Appendix B remains available.

The good Gram probability follows from the second line of (20). Use the original readout construction: truncate at $N\|v\|^2 \le K_0$, sample source indices with probabilities proportional to $|v_i|$, and use signed coefficients of magnitude $\|v\|_1/m \le \sqrt{K_0}/m$. The last line of (20) gives output sampling error at most $r/16$ uniformly over the finite input domain except with probability η. The energy truncation also fails with probability at most η. Including density-test, Gram, and shared-sketch aborts at internal stages, the total abort probability is at most $4L\eta$.

For each fixed input, glue the conditional transports as in Appendix B: retain the full original-weight marginal Q and the canonical outgoing prefix/future-weight marginal at each transition. Extend the transition by the previous conditional law of past original weights. This preserves every earlier computed prefix. The final candidate output is exactly the original network under the postactivation and readout injections defined by its rounded virtual computation. No robustness of an intermediate candidate is assumed.

On the non-aborted paths, Markov's inequality gives

$$\mathbb{P}\left\{\sum_{s<L}\|\eta^s\|^2/N > r^2/16, \text{ no abort}\right\} \le \frac{656Le}{r^2} = 0.656\alpha.$$

Readout rounding adds at most $\sqrt{e}/4 \le r/64$. Thus, outside probability at most $0.656\alpha + 4L\eta < \alpha$, the total disturbance satisfies

$$\|\eta\|_* \le \sqrt{r^2/16 + (r/16 + r/64)^2} = \frac{\sqrt{281}}{64} r < r/2.$$

The construction above does not require $m \le N$. Source sampling is with replacement, and duplicated indices give duplicated stored neurons. The Gram lemma also allows this: on its good event D has orthonormal columns, and $\ell \le q < m$. Hence the only width cutoff needed for this branch is $N > n_0$, which pays for the accumulated message exponents.

For $N \le n_0$, the original density bound implies a KL bound NK/t^2. The Gaussian entropy-moment inequality gives, for all physical standardized coordinates together,

$$\mathbb{E}_Q\|tZ_{\text{all}}\|^2 \le 4n_0K + 2P_{\text{small}}\log 2.$$

With probability at least $1-\alpha/4$, every actual coefficient is at most B_{small} in absolute value. On that event round the original network directly; otherwise use a zero-output default. The bounded rounded activations give the exact injection bound

$$\|\eta\|_*^2 \le \zeta^2[(d+1)^2 + (L-1)N^2] \le \zeta^2 L(n_0+d+1)^2 \le Le/16 < r^2/4.$$

Pad these small networks only to width n_0.

The large branch keeps the m selected rows at each layer. It stores $m(d+1)$ first-layer scalars, $2mk$ factor scalars for each internal matrix, and m readout scalars. The small branch stores at most P_{small} scalars. Every rounded scalar in either branch belongs to a fixed alphabet of at most $3 + 4B_{\text{code}}/\zeta$ values. The union of these two fixed finite classes, denoted $\mathcal{H}_{\text{fac}}$, therefore satisfies

$$\log|\mathcal{H}_{\text{fac}}| \le J_{\text{fac}}. \tag{35}$$

In particular one does not count the small branch as a dense network of width m.

We have proved that every law satisfying (29) admits a single candidate law Γ_Q on $\mathcal{H}_{\mathrm{fac}}$ with the following property: at each fixed input it has a coupling with $w \sim Q$ and a disturbance such that

$$h(x) = z_w(x;\eta), \qquad \|\eta\|_* < r/2 \tag{36}$$

except with probability at most α. Both marginals are independent of the input used to form the comparison coupling.

C.6 Posterior comparison and polynomial accounting

Proof of Theorem 7. The product target cylinder gives raw conditioning probability at least $\exp(-ND_*/t^2)$. The chi-square exponential moment for the readout, combined with this denominator, gives

$$Q_{S,N,t,c}\{E_v > B_v\} \le \exp\left(-\frac{NB_v}{4t^2} + \frac{N}{2}\log 2 + \frac{ND_*}{t^2}\right) \le \beta.$$

Let Q' be the posterior conditioned on $E_v \le B_v$. Unlike a generic law controlled only in relative entropy, it satisfies the literal density bound

$$\frac{dQ'}{d\mu_{N,t}} \le \frac{\exp(ND_*/t^2)}{1-\beta} \le \exp(NK/t^2).$$

Every original network in its support is robust on the common radius r. Consequently (36) implies clipped-score error at most $1/16$ on its good event.

Use the ramp ψ from the earlier statistical proof: it is one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and affine in between. Every training point has candidate ramp loss zero on the good coupling; every original test-margin failure has candidate loss one there. A relative Chernoff bound and a union bound over $\mathcal{H}_{\mathrm{fac}}$ give, on one sample event of probability at least $1-\delta$,

$$\mathbb{E}_D \psi(f(X)T(h(X))) \le \frac{2}{n}\sum_{i=1}^{n} \psi(f(X_i)T(h(X_i))) + \frac{8[J_{\mathrm{fac}} + \log(1/\delta)]}{n} \quad (h \in \mathcal{H}_{\mathrm{fac}}).$$

Averaging this inequality under $\Gamma_{Q'}$, applying the two marginal comparisons, and restoring the discarded readout tail bounds the desired risk by

$$\beta + 3\alpha + \frac{8[J_{\mathrm{fac}} + \log(1/\delta)]}{n} \le \varepsilon.$$

The finite class and sample event do not depend on N, t, which gives the simultaneous conclusion. Additional global clean-margin conditioning retains the same target cylinder, density bound, and readout-tail argument, so the same proof applies to that stated optional variant. □

Proof of Corollary 8. The product-cylinder constants give, at fixed ancillary resources,

$$K, B_v = O(CLF_0), \qquad r^{-2} = O(CL^2F_0), \qquad e^{-1} = O(CL^3F_0), \qquad \eta^{-1} = O(L).$$

The term $\log(1/\eta)$ in the rank bound is dominated by the target cost K. Therefore

$$q, V = O(C^2L^4F_0^2), \qquad p = O(CL^3F_0), \qquad m = O(C^3L^7F_0^3), \qquad n_0 = O(CL^4F_0).$$

The Gram and readout terms in the maximum defining m, including their logarithms, are dominated by its first term at this fixed-resource scale. It follows that

$$P_{\mathrm{fac}} = O(C^5 L^{12} F_0^5), \qquad P_{\mathrm{small}} = O(C^2 L^9 F_0^2).$$

All coefficient bounds and inverse meshes are polynomial in C, L, F_0, so $\log(3 + 4B_{\mathrm{code}}/\zeta) = O(\log(2+CL))$. Substitution into (24) and (25) proves the claimed order and both special cases. □

Remark 11 (A conservative joint polynomial for the factored code)**.** *Put*

$$S = 2 + d + L + C + c + c^{-1} + \varepsilon^{-1} + \log(1/\delta).$$

The target-cylinder estimates and (18) *give*

$$K, B_v = O(S^3), \qquad r^{-2} = O(S^6), \qquad e^{-1} = O(S^8), \qquad \eta^{-1} = O(S^2).$$

The new budgets consequently satisfy

$$q, V = O(S^{11}), \quad D_0 = O(S^{12}), \quad m = O(S^{19}), \quad n_0 = O(S^{13}), \quad p = O(S^8),$$

$$P_{\mathrm{fac}} = O(S^{31}), \quad P_{\mathrm{small}} = O(S^{27}), \quad B_{\mathrm{small}}^2 = O(S^{28}), \quad B_{\mathrm{code}} = O(S^{14}).$$

For the mesh, $A_f = O(S^{10})$ and $B_f = O(S)$; the latter uses the first lower bound on m to bound $\sqrt{q+1}/m$ by a constant. Thus $\zeta^{-1} = O(S^{44})$, where the factor-rounding denominator dominates the other displayed choices. It follows that

$$J_{\mathrm{fac}} = O(S^{31} \log S), \qquad n = O(S^{32} \log S) \subseteq O(S^{33}).$$

This degree-33 envelope is deliberately conservative. Like the refined width and depth powers above, it is a consequence of the present conventional argument and carries no Lean-certification claim.

References

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